SIMATS ENGINEERING

ASSESSMENT TOOL-1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

Duration: 30 Minutes

QUESTIONS: 20 **Total Marks:20**

Annexure A

MCQ Test

Code of Conduct:

I PRABHAVATHI R *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CO2_AT1_MCQTEST

Totalpoints20/20

NAME

PRABHAVATHI R

REGISTER NO

192321101

✓ Ininformedsearchstrategies,some algorithmsuseonlytheheuristic value todecidewhichnodetoexpandnext.Thisapproachfocusesonselecting the node that appears closest to the goal without considering the cost alreadyincurred.
Which algorithm follows this principle? 1/1

- ○ Breadth First Search
- ○ Uniform Cost Search
- ◉ GreedyBest-FirstSearch ✓
- ○ DepthFirstSearch

✓ A*searchiswidelyusedinpathfindingandgraphtraversalbecauseit combinestheactualcostfromthestartnodeandtheestimatedcosttothe goal. This balance allows it to find optimal solutions efficiently when the heuristic is admissible.
Whatis theevaluation functionused inA* search? 1/1

- ○ f(n)=g(n)
- ○ f(n)=h(n)
- ◉ f(n)=g(n)+h(n) ✓
- ○ f(n)=g(n)×h(n)

✓ Heuristic functions play a crucial role in informed search by guiding the algorithm toward the goal. A good heuristic should not overestimate the actual cost and should improve search efficiency.
Whatproperty ensures optimality in A* search? 1/1

- ○ Completeness
- ◉ Admissibility ✓
- ○ Depth limit
- ○ Backtracking

✓ Localsearch algorithmsoperate bystarting withan initialsolution and iterativelyimprovingitbasedonagivenobjectivefunction.Thesemethods are commonly used in optimization problems where the path is not importantbutthefinalsolutionis.
Whichof thefollowing is alocal search algorithm? 1/1

- ○ A* Search
- ◉ HillClimbing ✓
- ○ BreadthFirstSearch
- ○ UniformCostSearch

✓ In adversarial search, agents compete against each other in a game environment. The decision-making process involves considering both the agent's moves and the opponent's possible responses.

Which algorithm is commonly used for such decision-making? 1/1

- ○ Greedy Search
- ◉ MinimaxAlgorithm ✓
- ○ Backtracking
- ○ UniformSearch

✓ Adelivery robotin awarehouse selectspaths based onlyon theestimated distance to the destination without considering obstacles already encountered.Thissometimesleadstosuboptimalpathsbutfaster decisions. 1/1

Whichalgorithmis therobot using?

- ○ A*Search
- ◉ GreedyBest-FirstSearch ✓
- ○ Uniform Cost Search
- ○ Depth Limited Search

✓ AGPSnavigationsystemcalculatesroutesbycombiningthedistance already travelled and an estimate of the remaining distance to the destination.Thishelps infindingthe shortestpathefficiently. 1/1

Whichalgorithmbestrepresentsthisapproach?

- ○ DepthFirstSearch
- ◉ A* Search ✓
- ○ HillClimbing
- ○ Backtracking

✓ Auniversityisschedulingexamsformultiplesubjectswithconstraints such as no overlapping exams and limited rooms. The solution must satisfy all constraints while assigning time slots.
Thisproblemis bestmodeled as: 1/1

- ○ Game Playing Problem
- ◉ ConstraintSatisfactionProblem ✓
- ○ Heuristic Search Problem
- ○ OptimizationProblem

✓ AnAI-basedchessprogramevaluatesallpossiblemovesandanticipates theopponent'sresponsesbeforemakingadecision.Itassignsutility values to game states to determine the best move.
Which algorithm is being applied here? 1/1

- ○ GreedySearch
- ○ Hill Climbing
- ◉ MinimaxAlgorithm ✓
- ○ BreadthFirstSearch

✓ Arobotexploresanunknownenvironmentwhereitdoesnothaveprior knowledgeofobstacles.Itmakesdecisionsstep-by-stepandupdatesits knowledge as it moves.
Which type of agent is suitable for this scenario? 1/1

- ○ OfflineAgent
- ◉ OnlineSearchAgent ✓
- ○ Static Agent
- ○ Deterministic Agent

✓ A*search guarantees optimalsolutions only whencertain conditions are satisfiedbytheheuristicfunction.Iftheseconditionsareviolated,the algorithm may still find a solution but not necessarily the best one.
Why is an admissible heuristic important? 1/1

- ○ Itreducesmemoryusage It
- ◉ ensures optimality ✓
- ○ Itincreasesbranchingfactor
- ○ Itavoidsloops

✓ Inlargegametrees,evaluatingeverynodebecomescomputationally expensive.Alpha-Betapruninghelpseliminatebranchesthatdonotaffect the final decision, thereby improving efficiency.
Whatis the primarybenefit of Alpha-Betapruning? 1/1

- ○ Reducesdepthof tree
- ◉ Reducesnumberofnodesevaluated ✓
- ○ Changes game rules
- ○ Improvesheuristicaccuracy

✓ HillClimbingalgorithmcontinuouslymovestowardbetterstatesbasedon localimprovements.However,itmayfailtoreachtheglobaloptimumdue to certain limitations in the search space.
Whatisthemain drawbackofHill Climbing? 1/1

- ○ Highmemoryusage Revisiting
- ○ nodes
- ◉ Gettingstuckinlocalmaxima ✓
- ○ Requirescompletetree

✓ ConstraintSatisfactionProblemsinvolvevariables,domains,and constraints.Avalidsolutionmustassignvaluestoallvariablessuchthat no constraints are violated.
WhatensuresavalidCSPsolution? 1/1

- ○ Heuristicfunction
- ○ Path cost
- ◉ Constraintsatisfaction ✓
- ○ Depth-firsttraversal

✓ Inadversarialgames,evaluationfunctionsareusedwhenit isnotfeasible toexploretheentiregametree.Thesefunctionsestimatethedesirabilityof a game state.
Whatistheroleofanevaluationfunction? 1/1

- ○ Generate successors
- ◉ Estimateutilityofastate ✓
- ○ Reducebranchingfactor
- ○ Storemoves

✓ InA*search, theevaluation functionis calculatedusing boththe pathcost andheuristicvalue.Supposeanodehasapathcostof7andaheuristic estimate of 5.
Whatis thevalue of f(n)? 1/1

- (●) 12 ✓
- ( ) 2
- ( ) 35
- ( ) 7

✓ Aheuristicfunctionsometimesoverestimatestheactualcosttoreachthe goal.Thisviolatestheadmissibilityconditionrequiredforoptimalsolutions in A* search.
What happens in such a case? 1/1

- ( ) A*remainsoptimal
- (●) A*becomesnon-optimal ✓
- ( ) A* becomes BFS
- ( ) A*stopsworking

✓ Considerasearchtreewithabranchingfactorof2anddepthof4.The total number of nodes generated in a full binary tree can be calculated analytically.
Whatistheapproximatenumberofnodes? 1/1

- ○ 10
- ○ 15
- ◉ 31 ✓
- ○ 16

✓ Inaminimax tree,each nodehas 3childrenand thedepth ofthe treeis 3. Thetotalnumberofleafnodesdependsonthebranchingfactorraisedto the depth.
Howmanyleaf nodesare present? 1/1

- ○ 9
- ◉ 27 ✓
- ○ 6
- ○ 12

✓ Alpha-Betapruningimprovestheefficiencyofminimaxbyreducingthe number of nodes evaluated. In the best-case scenario, it significantly lowers time complexity.
What is the best-case time complexity? 1/1

- O(b^d)
- O(b^(d/2)) ✓
- O(d^b)
- O(log b)

ThisformwascreatedinsideofSaveethaInstituteofMedicalandTechnicalSciences.-Contactformowner Does this form look suspicious? Report